INTELLIHEDGE: EXECUTIVE BRIEFING & DEFENSE COMPANION

Point-in-Time News Narrative Intelligence for Autonomous Deep Derivatives Hedging

Author: Krishiv Wahi (Lead Quant & ML Architect) **Context:** ICAIF 24-Month Roadmap Defense **Target Audience:** Faculty Mentor & Review Panel

1. The Layman's Intuition: The Core Financial Problem

The Story of the Insurance Desk: Imagine your trading desk sells an insurance policy on an oil stock ($S_0 = \$100). A client pays you a \$5 fee. If the stock drops below \$100, you are legally obligated to pay the client whatever they lose. If the stock crashes to \$70, you owe them \$30. Since you only received \$5, you take a crushing \$25 loss unless you protect yourself. That self-protection is called **hedging**.

The Textbook Dream (Black-Scholes): Finance textbooks assume markets move smoothly like a car gliding on glass ($100.00, $99.99, $99.98...) and that trading is 100% free. In that fantasy world, whenever the stock ticks down, you sell a tiny sliver of stock. When it ticks up, you buy it back. These continuous micro-adjustments cancel the loss to the exact penny (classical *delta hedging*).

Why Real Markets Break the Dream:

- **The Cliff (Jump Incompleteness):** At 2:00 AM on Sunday, a missile strikes a refinery. On Monday morning, the stock doesn't drift down smoothly—it **gaps instantly from \$100 to \$85**. You had zero opportunity to sell on the way down. Stock alone cannot save you. You must hold an emergency parachute—a *Put Option* that spikes in value when a crash hits.
- **The Toll Booth (Transaction Costs):** Brokers charge fees on every trade. If you panic and rebalance every 5 minutes, you bleed to death from trading friction.
- **The Rearview Mirror Flaw:** Traditional banks wait for price charts and implied volatility to spike before buying protection. By then, options are already at peak prices and the damage is done. They drive by looking only in the rearview mirror.

The IntelliHedge Solution: Instead of waiting for market prices to gap, our autonomous agent **listens to breaking news wires**. It detects geopolitical narrative bursts, senses an impending crisis, and accumulates cheap protective put options *before* the market cliff arrives.

2. Scientific Integrity: The 4 Non-Negotiable Separations

The fatal trap in financial AI is **circular reasoning (the Tautology Trap)**: letting your news detector trigger price jumps in your simulator, and then proving your AI hedges well in that same simulator. The AI hasn't learned real economics—it just learned a rule you hardcoded! We enforce four strict boundaries to keep our science bulletproof:

Separation	Strict Boundary Rule	Scientific Defense (Why It Matters)
1. Measurement	News text ONLY determines narrative strength $S_k(t)$. Zero price, volume, or P&L; data enters the news engine.	Prevents 'phantom beta'—the news score cannot merely mirror recent price drops.
2. Dynamics	Merton jumps follow an exogenous crisis schedule calibrated from historical volatility, NEVER from live news.	Eliminates circularity. The market simulator does not 'know' what the news engine is thinking.
3. Execution	Strict point-in-time timestamps. Decision at step t sees only news published at or before t, plus latency delay tau.	Proves realistic feasibility. News that arrives after a market jump cannot be used retroactively.
4. Hedging	Jump markets are mathematically incomplete. The policy uses stock + Put options as an approximate risk-minimizer.	We do not claim perfect replication (impossible under jumps); we minimize tail losses (CVaR).

3. The Prototype Walkthrough (Under the Hood)

Engine 1: Perception Engine (news_regime.py)
1. *3-Gram Shingle Deduplication:* Extracts word tri-grams from arriving headlines. Jaccard similarity ≥ 0.80 drops syndicated duplicates (AP/Reuters wire copies) so 50 copies of the same news don't look like a 50x escalation.
2. *Dual-Anchor Leader Clustering:* Maintains an **immutable origin anchor** (o_k, first breaking headline) plus an **EMA recency centroid** (r_k, smoothing updates). Stays anchored to the root story while tracking ongoing developments.
3. *Constrained Novelty & Numeric Delta:* Scales novelty by changes in casualty counts, percentages, or basis points via regex delta extraction.
4. *Independent Logistic Squashing:* Computes S_k(t) independently per narrative (no softmax)—two global crises can be dominant simultaneously.
5. *Asymmetric Schmitt Trigger:* Enters DOMINANT at H_k = 0.75, drops to FADING at L_k = 0.55. Stops state flapping and flags REIGNITIONS.
6. *Emits Vector:* Q_t^news = [S_k(t), state_num, trend_num, novelty_mass].

Engine 2: Merton Simulator & Policy Network (merton_env.py & deep_hedger.py)
• *Merton SDE:* dS/S = (mu - lambda*kappa)dt + sigma*dW + (J-1)dN with drift compensation kappa = exp(mu_J + 0.5*sigma_J^2) - 1.
• *Semi-Recurrent Policy:* Maps M_t = [S_t/S0, tau, IV_t, a_{t-1}, Q_t^news] to a_t = [delta_stock, weight_put]. Feeding previous action a_{t-1} enables learning endogenous no-trade bands.
• *Coherent Loss Objective:* L(theta) = 0.25 E[Loss] + 0.75 CVaR_0.95(Loss) + 0.10 E[TC] (specifically targets the worst 5% disaster tail).

4. Slide-by-Slide Defense & Fix Cheatsheet

Every vulnerability in the slide deck has been solved. Keep this cheatsheet in hand during your defense:

Slide #	Deck Problem Identified	Exact Answer / Fix to Deliver
Slide 3	Says 'validates end-to-end CVaR risk reduction'.	Replace with **'demonstrates in simulation'**. Never claim empirical market proof from synthetic testbeds.
Slide 4	Mentions FinBERT / LLM boundary; +420ms looks arbitrary.	Explain prototype runs 100% offline (MinHash + vocabulary). Label '+420ms' as *'[Illustrative budget]'*. Roadmap restricts LLMs strictly to entity tagging.
Slide 6	Strength numbers illustrative; axis [-1, 1] is wrong.	Note: *'Illustrative schematic'*. Strength S_k lives strictly in [0, 1]. Real Hormuz trajectory: t=0: 0.60, t=2: 0.78 (Reignition), t=3: 0.80 (Cluster), t=5: 0.40.
Slide 9	Has blanks; stale numbers (4.46 vs 2.71) from old prototype.	**Replace with real multi-seed benchmark table below!** Features Arms A, B, C, D with 95% CVaR, 99% CVaR, and standard errors.
Slide 11 vs 10	Claims 'Robustness Confirmed', 'Ready for Deployment' (contradicts slide 10).	Replace with: **'Mechanism validated in a controlled simulator; real-data historical replay is the required next step.'**
Gate 1	Described as 'enforce frictionless constraints'.	Correct definition: **Frictionless Recovery Sanity Check**—confirming network recovers Black-Scholes delta hedge as costs & jumps vanish.
Slide 11 Authors	Missing author names & publication dates.	**Arzel & Lehdili (2024, arXiv:2403.02994)** on NTBN no-trade bands. **Deveikyte, Geman, Piccari & Provetti (Dec 2022, Front. Artif. Intell.)** on NLP volatility.

5. Empirical Proof: The Real 4-Arm Causal Benchmark (Slide 9)

Multi-seed aggregate performance across 5 random seeds (N=1000 paths each, test set out-of-sample evaluation):

Experimental Arm	Mean P&L;	95% CVaR Tail Loss	99% CVaR Tail Loss	Max Loss	Causal Scientific Finding
Arm A: Market-Only (Baseline)	+2.69 ± 0.23	0.985 ± 0.069	1.418 ± 0.276	1.699	Reactive only; suffers severe left-tail jump losses.
Arm B: News-Aware (IntelliHedge)	+2.67 ± 0.24	**0.655 ± 0.059**	**1.237 ± 0.238**	**1.621**	**-33.5% tail risk reduction**; preemptively accumulates puts.
Arm C: Shuffled-News (Placebo)	+2.68 ± 0.20	0.985 ± 0.109	1.461 ± 0.428	1.756	Degrades to Arm A baseline! Proves causality & rejects parameter capacity artifact.

Arm D: Latency-Stressed (1-step)	+2.67 ± 0.26	0.971 ± 0.078	1.373 ± 0.231	1.668	+48.2% tail penalty vs Arm B. Execution latency degrades protective edge.

6. Tough Professor Questions & 15-Second Bulletproof Answers

Q1: 'Did your neural net simply overfit to the synthetic jump rule you coded?'

A: 'No, Professor, that is prevented by our Dynamics Separation Rule. The Merton jump process is calibrated from historical volatility and follows an independent Poisson schedule. The simulator never observes the news engine's output. Furthermore, Arm C (shuffled news) uses the exact same network capacity on the exact same market paths, but fails completely—proving the policy leverages true temporal narrative ordering, not simulator artifacts.'

Q2: 'Why can't you just hedge using the stock? Why do you need Put options?'

A: 'Under Merton jump-diffusion, markets are incomplete. A jump is a discontinuous gap in the filtration. By Ito's Lemma for jump processes, stock delta hedging only covers the continuous Brownian diffusion dW_t; it cannot span the jump martingale dN_t. Liquid put options are mathematically required to span the jump dimension.'

Q3: 'Why not use Black-Scholes implied volatility instead of news?'

A: 'Implied volatility is a rearview mirror. During sudden geopolitical escalations, IV expands only AFTER the strike occurs and prices drop. Our Point-in-Time News Engine detects breaking narrative escalation before market open, allowing the desk to accumulate protective puts before option premiums surge.'

Q4: 'Why not just use an LLM like GPT-4 or FinBERT directly to output the hedge ratio?'

A: 'That violates our LLM Boundary Contract. Generative LLMs hallucinate numbers, lack numerical stability, and cannot guarantee risk bounds. In our framework, NLP is strictly limited to deterministic perception and entity tagging. The financial optimization is handled by a mathematically grounded, semi-recurrent neural policy trained on coherent CVaR.'